

## Lista de Relatividade Restrita:

① a)  $ds^2 = -c^2 dt^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2$

$$ds^2 = -c^2 d\tau^2$$

Traj 1  $\rightarrow \Delta Y = \int_{t_0}^{t_1} \sqrt{\frac{-dx^\mu \eta_{\mu\nu} dx^\nu}{c^2 dt}} dt$

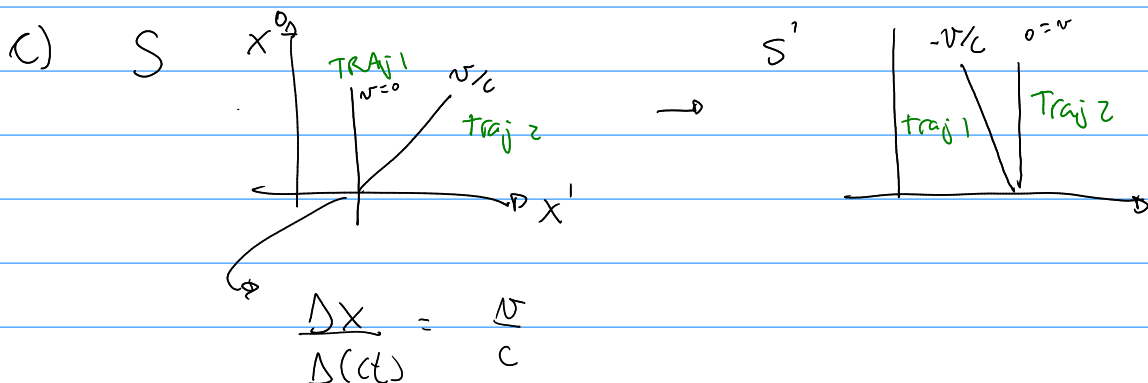
$$\Delta Y = \int_{t_0}^{t_1} \sqrt{\frac{-1}{c^2} c^2 dt^2} dt = \Delta \tau \quad \Rightarrow \boxed{\Delta Y = \Delta \tau}$$

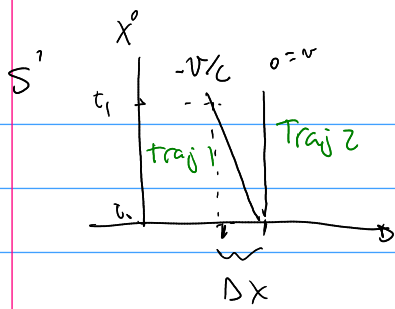
Traj 2  $\rightarrow \Delta Y = \int_{t_0}^{t_1} \sqrt{\frac{-1}{c^2} \left( -c^2 + \overbrace{v^2}^{(dx^1/dt)^2} \right)} dt$

$$\Delta Y = \int_{t_0}^{t_1} \sqrt{\left( 1 - \frac{v^2}{c^2} \right)} dt \Rightarrow \boxed{\Delta Y = \frac{1}{\gamma} \Delta t}$$

b)  $\Delta Y_1 = \Delta t_1$  e  $\Delta Y_2 = \frac{1}{\gamma} \Delta t_2$  (tempo próprio é invariante)

logo  $\Delta t_1 = \frac{1}{\gamma} \Delta t_2 \rightarrow$  Dil. Temporal





$$\rightarrow \Delta y_2 = \Delta t_2$$

$$\Delta y_2^2 = \Delta t_1^2 - \Delta x^2$$

$$\Delta y_2 = \sqrt{\left(1 - \frac{v^2}{c^2}\right)} \Delta t_1 \Rightarrow \Delta y_2 = \frac{1}{\gamma} \Delta t_1$$

logo:

$$\boxed{\Delta y_2 = \frac{1}{\gamma} \Delta t_1} \quad \text{X}$$

$$\frac{1}{\gamma} \Delta t_1 = \Delta t_2$$

$$\boxed{\Delta y_2 = \Delta t_2}$$

② a)

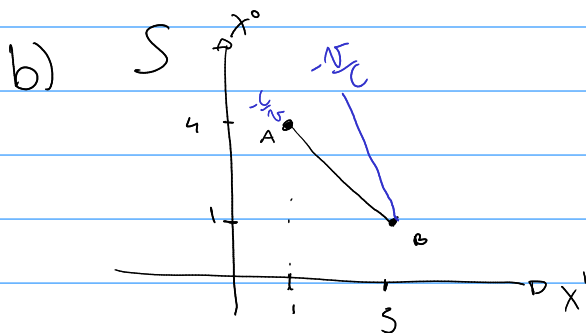
$$\left. \begin{array}{l} X_A^N = (4, 1) \\ X_B^N = (1, 5) \end{array} \right\}$$

$$\Delta X^N = X_B^N - X_A^N = (-3, 4)$$

$$\Delta S^2 = (-3 \ 4) \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} (-3 \ 4) = \begin{pmatrix} 3 \\ 4 \end{pmatrix} (-3 \ 4) = -9 + 16 = 7$$

logo

$$\Delta S = \sqrt{7} > 0 \quad (\text{intervalo tipo espaco})$$



$$\frac{-v}{c} = -\frac{(1-4)}{(5-1)} \Rightarrow \boxed{v = -\frac{3}{4}c}$$

c)  $S \rightarrow$  evento a

$S' \rightarrow$  indistinguido

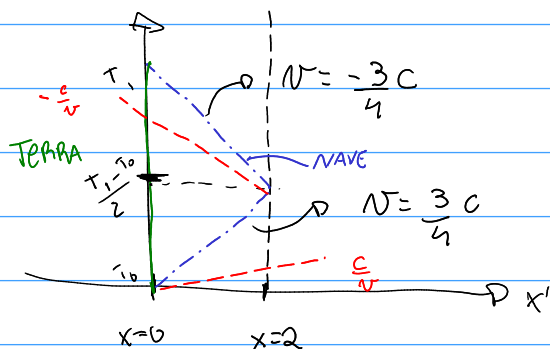
NÃO HÁ causalidade para intervalos tipo espaco.

9A

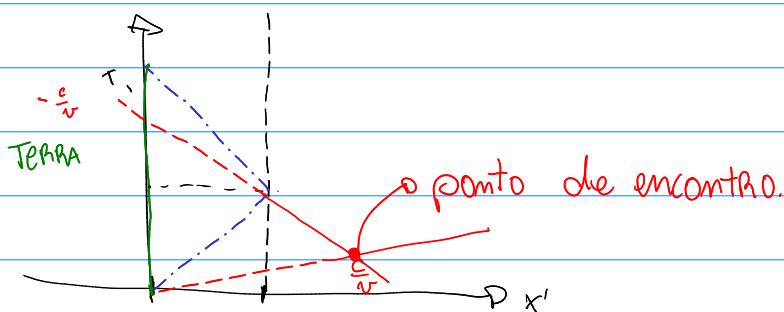
d)  $v$  menor  $\rightarrow$  P1  $v=0 \Rightarrow$  O B vem primeiro  
 $v$  maior  $\rightarrow$  P1  $v \approx c \Rightarrow$  O A vem primeiro

(3)

a, b)

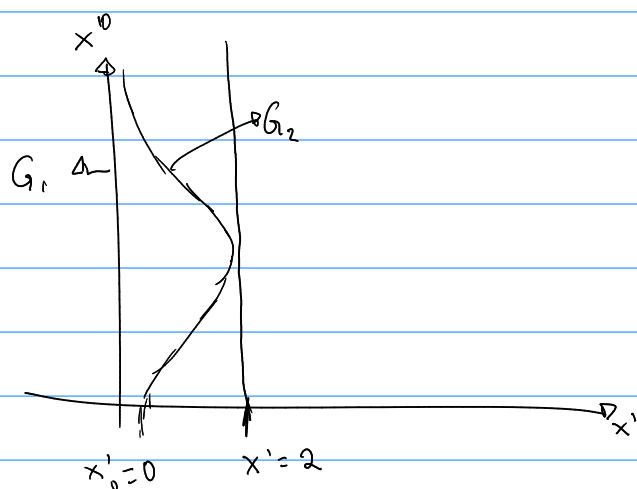


c)



R:

d)



Usaremos  $\frac{v}{c} = \tanh\left(\frac{x''}{x''_0}\right)$

O mapa é:

$$y' = x' - \sqrt{1 - \tanh^2\left(\frac{x^0}{x_*^0}\right)}$$

$$\begin{aligned} x^0 \rightarrow \infty &\Rightarrow y' = x' \\ x^0 \rightarrow 0 &\Rightarrow y' = x' \end{aligned}$$

$$y' = x' - \frac{d}{\cosh\left(\frac{x^0}{x_*^0}\right)}$$

pois  $\delta = \frac{1}{\sqrt{1 - \tanh^2\left(\frac{x^0}{x_*^0}\right)}}$

$$\text{Del} = \frac{L_0}{\delta} \Rightarrow y' = x' - \frac{d}{\delta}$$

$$\delta = \frac{1}{\text{sech}\left(\frac{x^0}{x_*^0}\right)} = \cosh\left(\frac{x^0}{x_*^0}\right)$$

Nosso MAPA é:

$$\phi_y^N(x) = (x^0, x' - \frac{d}{\cosh\left(\frac{x^0}{x_*^0}\right)}, x^2, x^3)$$

↙ inverso

$$x^0 = y^0$$

$$\phi_x^N(y) = (y^0, y' + \frac{d}{\cosh\left(\frac{y^0}{x_*^0}\right)}, y^2, y^3)$$

$$\Lambda^N_\alpha = \frac{\partial \phi_x^N}{\partial y^\alpha} =$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ -\frac{\tanh\left(\frac{y^0}{x_*^0}\right) d}{\cosh\left(\frac{y^0}{x_*^0}\right) x_*^0} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Agora,  $g_{\alpha\beta} = \frac{\partial \phi^\nu}{\partial y^\alpha} \eta_{\nu\mu} \frac{\partial \phi^\mu}{\partial y^\beta}$

$$g_{\alpha\beta} = \begin{pmatrix} 1 & \frac{-\gamma_0}{ch} \frac{d}{x_0^*} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{-\gamma_0}{ch} \frac{d}{x_0^*} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$g_{\alpha\beta} = \begin{pmatrix} -1 & \frac{-\gamma_0 d}{ch x_0^*} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ \frac{-\gamma_0 d}{ch x_0^*} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

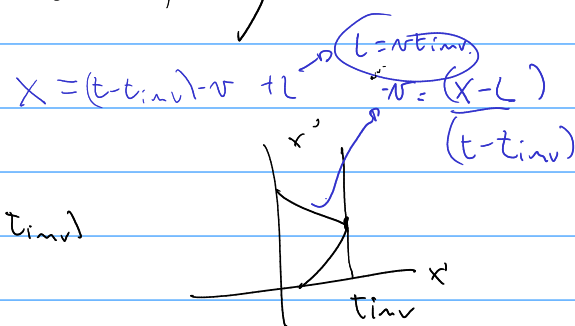
$$g_{\alpha\beta} = \begin{pmatrix} -\left(1 + \frac{\gamma_0^2}{ch^2} \frac{d^2}{x_0^{*2}}\right) & \frac{-\gamma_0 d}{ch x_0^*} & 0 & 0 \\ \frac{-\gamma_0 d}{ch x_0^*} & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \#$$

e)

$$x' = vt - 2v(t - t_{inv}) \Theta(t - t_{inv})$$

\* p1  $t < t_{inv} \Rightarrow \Theta = 0$  e  $x' = vt$

\* p1  $t > t_{inv} \Rightarrow \Theta = 1$  e  $x' = vt - 2vt + 2vt_{inv}$   
 $x' = -vt + 2vt_{inv}$



Assim:  $X^\mu = (x^0, v x^0 - 2v(x^0 - x_{inv}^0) \theta(x^0 - x_{inv}^0), x^2, x^3)$

$$\frac{dx^\mu}{dx^0} = v^\mu = (1, v - 2v \theta(x^0 - x_{inv}^0) - \underbrace{2v(x^0 - x_{inv}^0) \delta(x^0 - x_{inv}^0)}_{=0}, x^2, x^3)$$

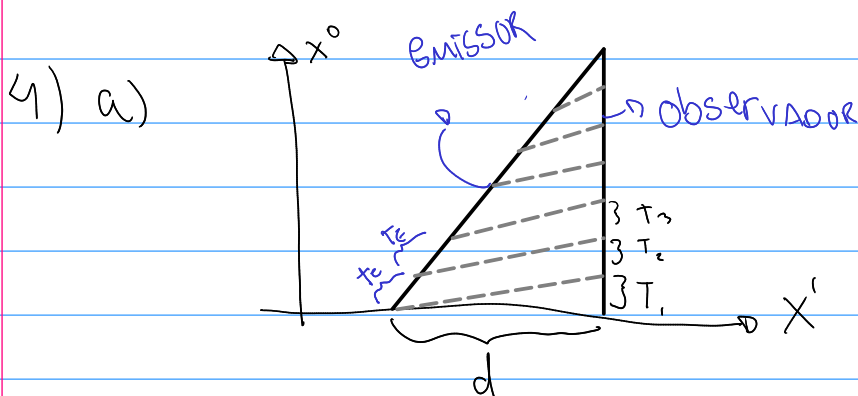
$$v^\mu = (1, v - 2v \theta(x^0 - x_{inv}^0), x^2, x^3)$$

$$\Delta y = \int_0^{x^0} \sqrt{\frac{dx^\mu}{dx^0} \eta_{\mu\nu} \frac{dx^\nu}{dx^0}} dx^0 = \int_0^{x^0} \sqrt{\left(\frac{dx^0}{dx^0}\right)^2 + \left(\frac{dx^i}{dx^0}\right)^2} dx^0$$

$$\Delta y = \int_0^{x^0} \sqrt{1 - [v - 2v \theta(x^0 - x_{inv}^0)]^2} dx^0 = \int_0^{x^0} \sqrt{1 - v^2 + 4v^2 \cancel{\theta(x^0 - x_{inv}^0)} - 4v^2 \cancel{\theta^2(x^0 - x_{inv}^0)}} dx^0$$

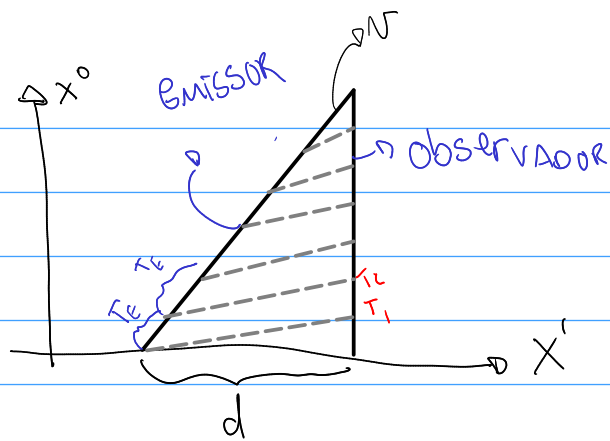
$$\Delta y = \int_0^{x^0} \sqrt{1 - v^2} dx^0 \Rightarrow \boxed{\Delta y = \frac{1}{\gamma} \Delta x^0}$$

f) a estrutura que rompe a simetria dos gêmeos é a aceleração e supor que a métrica é igual para os dois referenciais



essencialmente  $\Delta T_e \neq \Delta T_i$

$T_e \rightarrow$  tempo de cada pulso p/ emissor

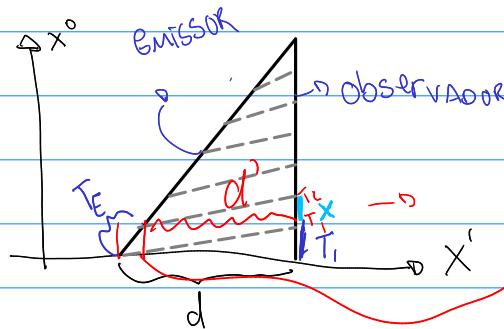
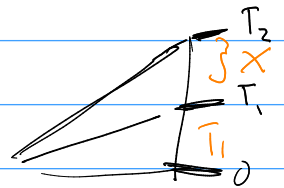


$$\Delta T_{\text{emissão}} = T_2 - T_1$$

$\xrightarrow{\text{observ}}$        $\xrightarrow{\text{observ}}$

→  $T_1$  chega em  $T_1 = \frac{d}{c} = T_e$

→  $T_2$  chega em  $T_2 = X + T_1$ , onde  $X$  é:



$$\Delta t' = \frac{d'}{c}$$

e  $d' = d - v_e T_{e2}$

logo:

$$T_2 = \frac{d - v_e T_{e2}}{c} + \frac{d}{c} \Rightarrow T_2 = \frac{2d - v_e T_{e2}}{c}$$

Portanto:

$$\Delta T_{\text{obs}} = T_2 - T_1 = \frac{d - v_e T_{e2}}{c} + \frac{d}{c} - \frac{d}{c} = \frac{d - v_e T_{e2}}{c}$$

$$\Delta T_{\text{obs}} = \frac{d - v_e T_{e2}}{c}$$

Como  $\frac{d}{c} = T_{e1} \Rightarrow \Delta T_{\text{obs}} = T_{e1} - \frac{v_e}{c} T_{e2}$

Cassini:

$$\Delta T_{\text{obs}} = T_{E_1} - \frac{v_E}{c} T_{E_2}$$

Como  $T_{E_1} = T_{E_2} = \Delta T_E$

Relação com  
os tempos Coordenados

$$\Delta T_{\text{obs}} = \Delta T_E \left( 1 - \frac{v_E}{c} \right)$$

Como  $\Delta y = \frac{1}{\gamma} \Delta t \Rightarrow \Delta y_o = \frac{1}{\gamma} \Delta t_o \mid \Delta y_E = \frac{1}{\gamma} \Delta t_E \rightarrow \Delta y_E = \Delta t_E$

Rel. com o  
 $\Delta y$

$$\Delta y_{\text{obs}} = \gamma \Delta y_E \left( 1 - \frac{v_E}{c} \right)$$

$$\Delta y_{\text{obs}} = \Delta y_E \frac{\left( 1 - \frac{v_E}{c} \right)}{\sqrt{\left( 1 - \frac{v_E^2}{c^2} \right)}} = \Delta y_E \frac{\sqrt{\left( 1 - \frac{v_E}{c} \right)^2}}{\sqrt{1 - \frac{v_E^2}{c^2}}}$$

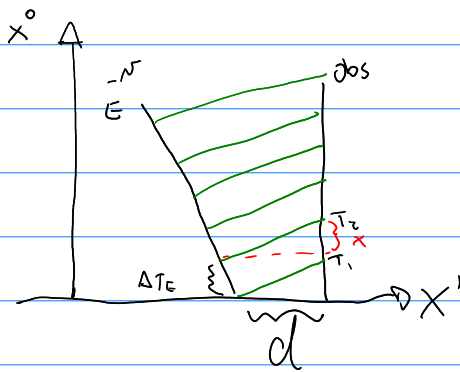
$$\Delta y_{\text{obs}} = \Delta y_E \frac{\sqrt{\left( 1 - \frac{v_E}{c} \right)^2}}{\sqrt{1 - \frac{v_E^2}{c^2}}} = \Delta y_E \frac{\left( 1 - \frac{v_E}{c} \right)^2}{1 - \frac{v_E^2}{c^2}} = \Delta y_E \sqrt{\frac{(1-\beta)(1-\beta)}{(1+\beta)(1-\beta)}} \quad \boxed{\beta = \frac{v_E}{c}}$$

$$\Delta y_{\text{obs}} = \Delta y_E \sqrt{\frac{1-\beta}{1+\beta}}$$

b)  $v_o = \frac{1}{\gamma_o}$  e  $v_e = \frac{1}{\gamma_e}$  logo:

$$\frac{1}{v_o} = \frac{1}{v_e} \sqrt{\frac{(1-\beta)}{(1+\beta)}} \Rightarrow \boxed{v_e = v_o \sqrt{\frac{1-\beta}{1+\beta}}} \quad \#$$

c) p/  $N = -v$



$$T_1 = \frac{d}{c}$$

$$T_2 = T_1 + x \quad \text{onde} \quad x = \frac{d'}{c} = \frac{d + v_e T_e}{c}$$

$$\Delta T_{\text{obs}} = T_2 - T_1 = \frac{d}{c} + \frac{d + v_e T_e}{c} - \frac{d}{c}$$

$$T_2 - T_1 = \frac{d + v_e T_e}{c} \Rightarrow \Delta T_{\text{obs}} = T_e + \frac{v_e T_e}{c}$$

Como  $T_1 = T_2 = \Delta T_e$

$$\Delta T_{\text{obs}} = \Delta T_e \left( 1 + \frac{v_e}{c} \right)$$

$$e \quad \Delta \gamma = \frac{1}{\gamma} \Delta t$$

$$\Delta \gamma_{\text{obs}} = \Delta \gamma_E \gamma \left( 1 + \frac{v_E}{c} \right) = \Delta \gamma_E \frac{\left( 1 + \frac{v_E}{c} \right)^2}{1 - \frac{v_E^2}{c^2}} = \Delta \gamma_E \sqrt{\frac{1+\beta}{1-\beta}} \quad \text{p/ } \beta = \frac{v_E}{c}$$

$$e \quad v_0 = \frac{1}{\gamma_0} \quad e \quad v_E = \frac{1}{\gamma_E}$$

$$\boxed{v_E = v_0 \sqrt{\frac{1+\beta}{1-\beta}}} \quad \#$$

(5) a)

Transfs de Lorentz:

$$x^1 = \gamma \left( x^1 - \frac{v}{c} x^0 \right)$$

$$e \quad x^{i'} = x^i \quad \text{p/ } i = 2, 3.$$

$$x^0 = \gamma \left( x^0 - \frac{v}{c} x^1 \right)$$

$$\text{Seja } x^\mu = (x^0, x^1, x^2, x^3) \quad e \quad x^{\mu'} = (x^{0'}, x^{1'}, x^{2'}, x^{3'})$$

$$x^{\mu'} = x^\nu \Lambda_{\nu}^{\mu'}$$

$$\begin{pmatrix} x^{0'} \\ x^{1'} \\ x^{2'} \\ x^{3'} \end{pmatrix} = \begin{pmatrix} x^0 & x^1 & x^2 & x^3 \end{pmatrix} \begin{pmatrix} A & B & C & D \\ E & F & G & H \\ I & J & K & L \\ M & N & O & P \end{pmatrix}$$

Logo, De cima:

$$X^{0'} = A X^0 + E X^1 + I X^2 + M X^3 = \gamma X^0 - \gamma \frac{v}{c} X^1$$

$$X^{1'} = B X^0 + F X^1 + J X^2 + N X^3 = \gamma X^1 - \gamma \frac{v}{c} X^0$$

$$X^{2'} = C X^0 + G X^1 + K X^2 + O X^3 = X^2$$

$$X^{3'} = D X^0 + H X^1 + L X^2 + P X^3 = X^3$$

Logo:

$$\rightarrow 0 = H, L, D, C, G, O, J, N, I, M$$

$$\rightarrow 1 = P, K$$

$$\rightarrow \gamma = A = F$$

$$\rightarrow -\gamma \frac{v}{c} = E = B$$

Logo:

$$\Lambda_{\mu\nu} = \begin{pmatrix} \gamma & -\gamma \frac{v}{c} & 0 & 0 \\ -\gamma \frac{v}{c} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

#

$$b) \Delta S^2 = -(\Delta x^0)^2 + (\Delta x^i)^2 \quad \text{p/ } i = 1, 2, 3$$

$$\Delta x^\mu = (\Delta x^0, \Delta x^1, \Delta x^2, \Delta x^3)$$

$$\Delta S^2 = (\Delta x^\mu)^\top \eta_{\mu\nu} (\Delta x^\nu) = (\Delta x^0 \Delta x^1 \Delta x^2 \Delta x^3) \begin{pmatrix} A & B & C & D \\ E & F & G & H \\ I & J & K & L \\ M & N & O & P \end{pmatrix} \begin{pmatrix} \Delta x^0 \\ \Delta x^1 \\ \Delta x^2 \\ \Delta x^3 \end{pmatrix}$$

Logo

$$-(\Delta x^0)^2 + (\Delta x^1)^2 + (\Delta x^2)^2 + (\Delta x^3)^2 = (\Delta x^0 \ \Delta x^1 \ \Delta x^2 \ \Delta x^3) \begin{pmatrix} A x^0 + B x^1 + C x^2 + D x^3 \\ E x^0 + F x^1 + G x^2 + M x^3 \\ I x^0 + J x^1 + K x^2 + L x^3 \\ M x^0 + N x^1 + O x^2 + P x^3 \end{pmatrix}$$

Chegamos em:

$$-(\Delta x^0)^2 = A x^0^2$$

$$(\Delta x^1)^2 = F \Delta x^2 \text{ e semelhante.}$$

O resto é zero.

Logo:

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

#

c) dado  $\Delta S = (\Delta x^\mu) \eta_{\mu\nu} (\Delta x^\nu)$  p/  $\Delta x^\mu \rightarrow \Delta x^{\mu'} = \Lambda^\mu_{\alpha} x^\alpha$

$$\Delta S = \bar{\Delta S} \text{ e } \bar{\Delta S} = (\Lambda^\mu_{\alpha} x^\alpha)^T \eta_{\mu\nu} \Lambda^\nu_{\rho} x^\rho$$

$$(\Delta x^\mu)^T \eta_{\mu\nu} (\Delta x^\nu) = (\Lambda^\mu_{\alpha} x^\alpha)^T \eta_{\mu\nu} \Lambda^\nu_{\rho} x^\rho$$

$$(\Delta x^\mu)^T \eta_{\mu\nu} (\Delta x^\nu) = (\Delta x^\alpha)^T (\Lambda^\mu_{\alpha})^T \eta_{\mu\nu} \Lambda^\nu_{\rho} x^\rho$$

Portanto:

$$\eta_{\alpha\rho} \equiv \Lambda^\mu_{\alpha} \eta_{\mu\nu} \Lambda^\nu_{\rho} \quad \#$$

⑥ a)

$$\frac{\partial f}{\partial v} = \lim_{\epsilon \rightarrow 0} \frac{f(x + \epsilon v) - f(x)}{\epsilon}$$

$$\frac{\partial f}{\partial v} = \lim_{\epsilon \rightarrow 0} \frac{f(x^1 + \epsilon, x^2 + \epsilon, x^3 + \epsilon) - f(x^1, x^2, x^3)}{\epsilon}$$

$$f(x + \epsilon) \approx f(x) + \left. \frac{\partial f}{\partial x} \right|_{\epsilon=0} \epsilon + \mathcal{O}(\epsilon^2) \Rightarrow f(x^0 + \epsilon) \approx f(x^0) + \epsilon \cdot \frac{\partial f}{\partial x^i} x^0$$

Così

$$f(x^1 + \epsilon, x^2 + \epsilon, x^3 + \epsilon) \approx f(x^1, x^2, x^3) + \epsilon \frac{\partial f}{\partial x^i} v^i$$

$$\frac{\partial f}{\partial v} = \lim_{\epsilon \rightarrow 0} \frac{\cancel{f(x^1, x^2, x^3)} + \epsilon \frac{\partial f}{\partial x^i} v^i - \cancel{f(x^1, x^2, x^3)}}{\epsilon} = v^i \frac{\partial f}{\partial x^i}$$

Così vale  $\forall f$ :

$$\left[ \frac{\partial}{\partial v} = v^i \frac{\partial}{\partial x^i} \right] \text{ logo } \left[ v \equiv v^i \frac{\partial}{\partial x^i} \right] \neq$$

b)  $y^\mu = \phi_y^\mu(x)$  con inversa  $x^\mu = \phi_x^\mu(y)$

in  $x$

in  $y$

$$v = v^\mu \frac{\partial}{\partial x^\mu} = v' = v'^\mu \frac{\partial}{\partial y^\mu}$$

De  $\phi_y^\mu(x) = y^\mu \Rightarrow \underbrace{v(y^\mu)}_{v^\mu \text{ ma scilicet } y = v^\mu} = v \phi_y^\mu(x) \quad v = v^\mu \frac{\partial}{\partial x^\mu}$

Portanto

$$v^{\mu'} = v^{\nu} \frac{\partial \phi^{\mu'}_{\gamma}(x)}{\partial x^{\nu}}$$

c)

$$\phi_{\gamma}(x) = y^{\mu}$$

Transf de Lorentz.  $\rightarrow x^{\mu'} = \Lambda^{\mu'}_{\alpha} x^{\alpha}$

Como

$$v^{\mu'} = \frac{\partial \phi^{\mu'}_{\gamma}(x)}{\partial x^{\nu}} v^{\nu}$$

e  $\frac{\partial \phi^{\mu'}_{\gamma}(x)}{\partial x^{\alpha}} = \Lambda^{\mu'}_{\alpha}$

Luego

$$v^{\mu'} = \Lambda^{\mu'}_{\nu} v^{\nu}$$

(7)

a)  $df = \frac{\partial f}{\partial x^i} dx^i$

e do enunciado

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \int_t^{t+\epsilon} \frac{\partial f}{\partial x^i} \frac{dx^i}{dt} dt = \lim_{\epsilon \rightarrow 0} \int_t^{t+\epsilon} \frac{df}{dt} dt = \lim_{\epsilon \rightarrow 0} \frac{f(x(t+\epsilon)) - f(x(t))}{\epsilon}$$

$$= \frac{df}{dt} = \frac{\partial f}{\partial x^i} \frac{dx^i}{dt} = \frac{\partial f}{\partial x^i} v^i = \left( \frac{\partial f}{\partial v} \right) \dot{x}.$$

b)  $w = \frac{\partial f}{\partial x^{\mu}} e^{\mu}$  e  $v = \frac{dx^i}{dt} \bar{e}_i$

$$w(v) = \frac{\partial f}{\partial x^{\mu}} e^{\mu} \left( \frac{dx^i}{dt} \bar{e}_i \right) = \frac{\partial f}{\partial x^{\mu}} \frac{dx^i}{dt} e^{\mu}(\bar{e}_i) = \frac{\partial f}{\partial x^{\mu}} \frac{dx^i}{dt} \delta^{\mu}_i = \frac{\partial f}{\partial x^{\mu}} \frac{dx^{\mu}}{dt}$$

logo  $w(v) = w_\mu v^\mu$

c)  $Y^\mu = \phi^\mu_Y(x)$  com  $J^\mu_\nu = \frac{\partial \phi^\mu_Y(x)}{\partial x^\nu}$

$d \equiv dx^i \wedge \frac{\partial}{\partial x^i}$   
 $dx^\mu = d\phi^\mu_X(x)$  e  $df = \frac{\partial f}{\partial x^\mu} dx^\mu$

$dx^\mu = dy^i \wedge \frac{\partial}{\partial y^i} \phi^\mu_X(y) \Rightarrow$

$dx^i = \frac{\partial \phi^\mu_X(x)}{\partial y^i} dy^i$

Agora como  $w = w_\mu dx^\mu = w'_\nu dy^\nu$

$w = w_\mu \frac{\partial \phi^\mu_X(x)}{\partial y^\nu} dy^\nu = w'_\nu dy^\nu$

logo:

$w'_\nu = w_\mu \frac{\partial \phi^\mu_X(x)}{\partial y^\nu}$

d)  $J^\mu_\nu = \frac{\partial \phi^\mu_X(x)}{\partial y^\nu}$

$x^\mu = \phi^\mu_X(y) = \phi^\mu_X(\phi_Y(x))$

$\frac{\partial x^\mu}{\partial x^\nu} = \frac{\partial \phi^\mu_X(\phi_Y(x))}{\partial \phi_Y(x)} \frac{\partial \phi_Y(x)}{\partial x^\nu}$

$\delta^\mu_\nu = \frac{\partial \phi^\mu_X}{\partial \phi_Y} \frac{\partial \phi_Y}{\partial x^\nu} \rightarrow \delta^\mu_\nu = J^\mu_\alpha J^{\alpha}_\nu$   
 $\Pi = J \cdot J^{-1}$